Blood pressure, sodium intake, and sodium related hormones in the Yanomamo Indians, a "no-salt" culture.

The Yanomamo Indians are an unacculturated tribe inhabiting the tropical equatorial rain forest of northern Brazil and southern Venezuela who do not use salt in their diet. The group therefore presented an unusual opportunity to study the hormonal regulation of sodium metabolism in a culture with life-long extreme restriction of dietary sodium, with parallel observations on blood pressure. Blood pressures increased from the first to second decade but, in constrast to civilized populations, do not systematically increase during subsequent years of life. In twenty-four hour urine collections on adult male Indians, excretion of sodium averaged only 1 plus or minus 1.5 (SD) mEq. Simultaneous plasma renin activities were elevated and comparable to those of civilized subjects placed for brief periods on 10 mEq sodium diets. Similarly, excretion rates of aldosterone equaled those of acculturated subjects on low sodium diets. The findings suggest that the hormonal adjustments to life-long low sodium intakes are similar to those achieved in acute sodium restriction of civilized man. Parenthetically, these elevated levels of aldosterone and renin were probably the norm for man during much of human evolution and suggest that the values observed in civilized controls are depressed by an excessive salt intake in contemporary diets.